

Project Overview

In today's world, where almost every messaging app tracks your personal information or ties your identity to a phone number or email, Xipher emerges as a privacy-first solution. It's a secure and decentralized mobile messaging app that lets users communicate completely anonymously — no phone number, no email, no central server involved.

Xipher is built using the Oxen framework, an open-source, privacy-focused platform designed for anonymous communication. While many apps rely on central servers, Oxen operates on a decentralized service node network, meaning your messages don't pass through any single authority that could monitor or store them. Every conversation is protected with end-to-end encryption, ensuring no one (not even the service provider) can read or track what you're saying or who you're talking to.

One of the standout features of Xipher is its local authentication system: you can lock the app with a custom PIN. So even if someone gets access to your phone, they can't open the app without your permission. We also stripped out non-essential features to keep the app lightweight, distraction-free, and focused solely on secure communication.

Instead of just being “another chat app,” Xipher is a real-world example of how secure, ethical, and open-source technology can fight back against modern surveillance. It's designed not just as a tech demo, but as a fully working application — ready to protect people's privacy.

Problem Statement

Most of today's messaging apps ask for your phone number, email, or other personal details before you can even start chatting. Behind the scenes, they often track your IP address, collect metadata, and store your chats on central servers — creating a digital trail that can be monitored or misused. Even with “encrypted” apps, your identity isn't really anonymous. In a time where digital privacy is constantly under threat, there's a serious need for a truly private, secure, and anonymous messaging platform — one that doesn't just promise security but is actually built from the ground up for it.

Aim

The aim of this project is to build Xipher a fully functional, secure, and decentralized mobile messaging app that gives users complete privacy and anonymity. The goal is to develop a platform where users can communicate freely without revealing their identity, without trusting any central authority, and without compromising on security or usability.

Objectives

- i. To develop a secure messaging application that does not require phone numbers or email addresses, enabling anonymous communication through randomly generated user IDs.
- ii. To implement strong end-to-end encryption protocols along with a local authentication mechanism (PIN) for enhanced data and device-level security.
- iii. To design a minimal, user-friendly interface that prioritizes usability, performance, and a distraction-free communication experience.

- iv. To apply core cybersecurity principles such as decentralized architecture, cryptographic security, and privacy-by-design throughout the development lifecycle.
- v. To demonstrate the feasibility of building a functional, privacy-focused communication platform using open-source frameworks and responsible development practices.